

PV, thermoelectric and functionally graded material layers.

Efficiency and cost

PV cell efficiency is the ratio of the electrical energy produced by a PV cell to the energy from sunlight falling on the cell. The typical efficiency of a commercial single-crystalline silicon solar cell is about 13 per cent at room temperature. However, the efficiency of a hybrid solar panel put under the irradiation of $750\text{W}/\text{m}^2$ with water flow is about 18 per cent. Moreover, hot water is available for domestic/commercial use. Without water flow the PV temperature reaches 90°C and its efficiency decreases to 13 per cent.

Typical electric power generation from a PV cell is $120\text{W}/\text{m}^2$. From bulk thermoelectric module it is $10.3\text{W}/\text{m}^2$. This increases to $27\text{W}/\text{m}^2$ when a quantum-well thermoelectric module is used, as the 'Z' value of the quantum-well thermoelectric module is twice that of the bulk thermoelectric module. The approximate cost of a hybrid panel using the PV-hot-water module is about Rs 20,000/ m^2 . PV-thermoelectric-hot-water panel costs about Rs 100,000/ m^2 .

Light-splitting trick

To create highly efficient solar cells, researchers have employed a novel engineering strategy where the solar power device splits light into high- and low-energy beams and routes these to different electricity producing materials. Different varieties of semiconductors layered in solar cells respond to photons of varying energies to produce electricity. Until now, however, the requirement of atomic structures of such layers line-up to allow proper crystal growth has limited the combinations that could be used to gain better efficiency.

The sunlight can be split into different wavelengths by using a prism. These wavelengths can be used separately for different electricity-producing solar cells. This was done in 1970 and the rainbow cell so cre-

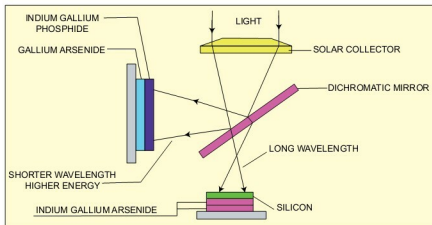


Fig. 3: Light-splitting trick

ated had a very low efficiency as a lot of sunlight was lost. However, recent advances in optics have made it possible to manufacture dichromatic mirrors.

Mirrors made with dichromatic material can process and split the light with a 93 per cent efficiency. Fig. 3 shows the schematic diagram of such a device. White incident sunlight is split with the help of dichromatic mirror into two components—shorter wavelength with higher energy, and longer wavelength with lower energy. While the shorter-wavelength light is utilised in indium-/gallium-based solar cells, the longer-wavelength light is used in silicon-based cells. Under laboratory condition, the efficiency has been reported to reach 40 per cent. The goal of the manufacturing cost has been set to about Rs 140 per watt of energy (without government subsidy).

Solar cells that work at night

A new breed of solar cells that harvest energy at night could double the output of conventional panels. It sounds like an oxymoron, but a new breed of nanoscale light-sensitive antennae could soon make this possible, heralding a novel form of renewable energy.

Under ideal conditions, solar arrays of billions of tiny antennae can collect 84 per cent of incoming pho-

tons. A complete system of this new breed of electronic solar cells would have an overall efficiency of 46 per cent considering all losses, whereas the overall efficiency of the most modern silicon solar cells is stalled at 25 per cent. What is more, if the orientation of the panel with respect to the sun is wrong, light reflects off the silicon solar cell instead of being absorbed to generate power. The key to these new devices is their ability to harvest radiation (infrared).

Beyond the red end of the visible range, but at frequencies higher than of radar waves and microwaves, is the infrared region of the electromagnetic spectrum between 10^{12} and 5×10^{14} Hz frequencies (or wavelengths from 0.1 to 7.5×10^{-5} centimetre). Nearly 50 per cent of the sun's radiant energy is emitted in the infrared region, with the rest is primarily in the visible region.

Infrared is remitted by the earth's surface after the sun goes down, meaning that these new-generation solar cell antennae can capture some energy even during the night. Unlike PV cells, which use photons to liberate electrons, the antennae resonate when hit by light waves, generating an alternating current that can be harnessed.

To build an array that could capture both visible and infrared radiations, multiple layers of antennae—with each layer tuned to a different optical frequency—are to be